People and technology

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All too often technological development has appeared to neglect the most complex component of any system — the human being! We tend to forget that communication is brain to brain. This issue introduces a range of work that is redressing this balance concerning the relationships between people and technology.

1. Introduction

Our standards for quality and acceptability are constantly evolving as technological capabilities increase. Thirty years ago, black and white TV pictures were the norm, with colour pictures a rare luxury. Now, when we see a repeat of a black and white TV programme, it seems almost comically dated and impoverished. The same trend in user expectations applies to information and communications technology (ICT) products and services. Acceptability standards for quality and reliability have evolved in line with improvements in networks. Because the change is progressive and gradual, at no point do we perceive a huge step in performance, but we would certainly notice if there were a sudden reversion to 1950s' standards of international telephony access, transmission quality and connectivity.

People are prepared to trade off quality of service against utility. For example, when mobile telephony was an exclusive novelty, the fact that neither transmission performance nor connectivity were as good as that experienced over a fixed link was accepted. However, as mobile telephony has become more of a mass-market phenomenon, peoples' expectations about the quality of service have evolved.

2. Benefits of considering the user — the 'science' of people

The complexity of emerging technologies makes it necessary to employ objective measurement criteria in order to arbitrate performance optimisation while ensuring that perceived performance is maximised. The introduction of compressive speech codecs, echo cancellers, and other nonlinear processing in voice-only networks, has already shown both the requirement for and benefit of objective quality metrics. In particular, BT's auditory model-based speech quality assessment system, PAMS, has provided vital end-to-end performance assessment for cross-network, cross-operator, and emerging voice over the Internet (VoIP) technology performance. In many cases network assessment and business-critical performance analysis is only possible using an appropriate objective metric, since subjective tests are impossible in field conditions. Perceptually motivated techniques are already providing an important business benefit across several parts of BT.

3. Perceptual modelling

Perceptual modelling is a powerful example of matching technology to people. This allows data coding and compression, as well as objective performance assessment, to be guided by what is perceived by the end user.

A common example of perceptually motivated coding is the MPEG family of algorithms for audio coding, as used in the new generation of DVD (digital versatile disk) players, and digital audio broadcasting. Here audio signals are compressed using algorithms that are designed to selectively discard information that the human auditory system could not detect. The MPEG audio format exploits perceptual phenomena such as upward and downward spread of masking in the frequency domain, whereby one sound can make another sound, that could be resolved in isolation, undetectable. This allows the signal to be compressed with a data reduction of fivefold (or more) with little or no effect on the perceived sound quality.

Papers by Antony Rix et al (p 24) and Mike Hollier et al (p 35) describe work that is under way at BT Laboratories to broaden our understanding of auditory, visual and multimodal perception. The objective is to allow us to do 'more for less' in terms of perceived quality of service (QoS) for a given bandwidth.

4. Technology for people

Human factors (or ergonomics) grew from the study of the 'knobs and dials' aspects of physical systems such as aircraft cockpits. There has been a tendency to associate the subject with superficial aspects of a system — for example the shape, size and colour of elements of a screen-based user interface. This perspective of the relationship between people and machines has led to the tendency to leave the end user until last, after all the 'real' technical work has been completed.

Fortunately for the users, the state of the art has moved on. Systems are commonly designed with the user's needs at the centre of things right from the outset. This approach, 'user-centred design', is exemplified in the paper by Kathy Pollard and Richard Blyth (p 69) which describes a project to substantially improve a Web-based user interface. This case study is one example of how BT products and services can now be designed around the needs of the user.

A good example of where technology has not fulfilled its popular promise is in the area of person-to-person videotelephony. Videoconferencing has certainly taken off strongly in a number of areas, and BT has a substantial and successful range of conferencing products in its portfolio. However, if we look at how the future was seen 10 or 20 years ago, futurologists predicted the widespread adoption of videotelephony, with voice-plus-video replacing the voice-only telephone.

Alwyn Lewis and Charles Nightingale's paper (p 47) discusses this paradox, and considers some ways in which a better understanding of people, and some new thinking about their relationship with video technology could be used to further increase the uptake.

Li-Qun Xu et al's paper (p 59) describes the latest research to improve multi-party videoconferencing. The work aims to make the people involved in such conferences feel closer, and more integrated in a coherent conference setting. Novel camera views and advanced processing are used to create perspective views of participants, to capture more of the facial expression and body movements that are so important to human communication.

Knowledge and information management is becoming an increasingly important topic in the business world. Developments such as the Internet and the corporate intranet allow users unparalleled access to data of all types from all over the globe. In this case, there is the potential for technology to drown people in information, rather than carefully matching and tailoring information to the individual. Research is under way at BT to develop tools to help with information filtering and matching. For example, Crossley et al's paper (p 76) describes 'Knowledge Gardens', a novel and innovative way for a group of people working together to visualise and manage shared information resources.

5. Real-world studies of people

While a great deal has been learnt about people and technology by carrying out careful experiments in the laboratory, controlled laboratory studies also have their limitations. We are therefore making increasing use of field studies to observe exactly how information and communications technologies are used in the real world, and to understand the context of use.

Ben Anderson et al (p 85) describe an ambitious and groundbreaking study of family life in the digital home. This involves a long-term (three-year) study of 600 households, giving BT an unprecedented appreciation of the ways in which different family members communicate, and, in particular, of the ways in which telephony and computers are used within the home.

A further field study, described by Karina Tracey et al (p 98), explores the ways in which ICT can play a role in supporting home-based learning, and in linking home and school. Again, an extended study is involved, carried out in a real community, to give a valuable insight into the ways in which parents, teachers, and children might use ICT to support learning and the wider educational process.

The process of introducing a new technology and a new way of working is a classic example of 'management of change'. A degree of trust between employees and managers, or between different groups of stakeholders is essential. Diane Warne and Chris Holland (p 111) describe a model of 'trust' which can be applied to provide early warning of issues in the design and introduction of a wide range of new systems. Their paper describes a model of flexible working in a BT software engineering centre where issues affecting trust were handled well. The model has also been applied in the past to the financial sector, and will be used extensively in looking at the trust issues associated with electronic commerce.

6. People in virtual environments

We have described above the importance of studying people in real-world settings. In contrast, technology now offers us the opportunity to experience shared 'virtual' environments. The October 1997 edition of this Journal has already surveyed this general area of work in some detail. This issue contains some important new research directions that are being followed by some of our university collaborators.

Mel Slater (p 120) from UCL reports a recent series of studies into the effects that virtual environments have on real people's behaviours. This research demonstrates that the technical characteristics of the virtual environment exert a strong influence on perceived status in a virtual world. A further study in the same paper suggests that people can react to a virtual audience in much the same way as they react to a real one.

Roy Kalawsky et al (p 128) describe work carried out at Loughborough University for the scientific evaluation of virtual environments. The discipline of human factors has generated a range of tools and techniques for evaluating users' reactions to more traditional products and services, such as telephones and desktop multimedia systems. The work reported in this paper describes new research to develop the extra tools and techniques needed to make principled design decisions about the next generation of virtual reality systems.

7. Dialogue engineering

There has recently been a rapid increase in the deployment of interactive voice response (IVR) services. It is now an everyday experience for most people to interact with voice messaging systems, 'automated attendant' systems (often as part of a customer service operation), and IVR systems for services such as home banking and home shopping. It is also becoming more common for people to encounter speech recognition technology in addition to the more established dual tone multifrequency (DTMF) controlled services.

A great deal of work has been carried out by BT to improve the match between speech technology and people.

Fred Stentiford and Pam Popay (p 142) survey the range of work aimed at increasing the usability of IVR services using tools and techniques such as the BT Dialogue Style Guide, and trials and evaluations of new IVR services prior to launch.

Fergus McInnes et al (p 160), from the CCIR at Edinburgh University, describe an experiment examining the effect of different prompts on users' responses to an IVR banking system. The research demonstrates that users' responses tend to be longer if system prompts are more verbose — an interesting example where we can use our understanding of dialogue engineering to subtly tailor the users' responses to the needs and capabilities of the system, a case of gently matching people to technology.

It can be very daunting for a user who encounters a complex and poorly designed IVR service. It can be easy to get 'lost' in a service if there is no clear understanding of the overall design and structure that has been used. One technique that can be employed to help the user is to use an everyday metaphor for the service that is being offered.

Rachael Dutton et al (p 172) report an experiment on a home shopping service, and examine the benefits offered by metaphors based on a department store and a magazine. These allowed users to bring their existing real-world knowledge and experience to bear in the context of the otherwise unfamiliar IVR service. The technology is therefore matched to the users' models of how they go about shopping for items in everyday life.

Early speech-recognition-based systems were required to recognise just a few words and phrases. This can be accomplished using relatively simple dialogue structures. As the vocabulary of items which the user can input increases, the complexity and potential for user confusion increases dramatically. David Attwater and Steve Whittaker (p 149) describe recent work to research new dialogue design principles to cope with much larger input vocabularies.

Further developments in allowing speech recognition systems to cope with the complexity and variety of peoples' natural fluent speech are described in the final paper on dialogue engineering by David Attwater et al (p 178). This work offers a bridge between the constraints of traditional IVR systems and a new era in which speech technology is properly matched to people, allowing natural and fluent human/machine communication.

8. Thinking for the future

This issue starts with a stimulating paper by Alwyn Lewis (p 15), who describes some of the new ways of creative thinking that are needed to engineer future products and services. The increasing pace of change, and the rapid emergence and large-scale adoption of novel technologies, such as the World Wide Web, mean that we have to embrace revolution as well as steady evolution in our thinking.

9. Conclusions

On reading this issue, you will become aware of the broad area it surveys. There is great scope for a better match between people and technology, and a range of initiatives are under way, both within BT Laboratories and in the university community, to move in this direction. We hope you enjoy the papers in this issue, and are encouraged to follow up particular directions of interest with the authors.

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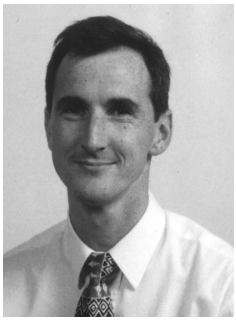
He joined the Human Factors unit at BT Laboratories (BTL) in 1985, and worked on colour graphic displays for network management.

He then went on to lead a team working to improve the usability of BT's wide range of products and services.

He currently heads the Human Factors Research unit at BTL and jointly leads the Psy-Tech research initiative investigating the links between people and technology.



Mike Hollier obtained a BEng (Hons) degree in Mechanical Engineering from Plymouth Polytechnic in 1987. After working as a hi-fi loudspeaker designer, he joined BT Laboratories to work on a variety of acoustics projects including the design of a novel noise-cancelling handset.

He obtained an IOA diploma in acoustics in 1989, receiving an industry award for his research into the vibrational behaviour of light structures. He received his PhD from the University of Essex in 1995 and since 1996 has led the Multi-modal Perception group with projects on auditory, visual and multi-sensory perception.

He is a visiting lecturer to the University of Essex, where he is a Fellow of the Department of Electronic Systems Engineering. He is a Chartered Engineer and a Member of the IOA and AES.

Fred Stentiford studied Mathematics at St Catherine's College, Cambridge, and obtained a PhD in Computer Aided Design at Southampton. He first joined the Plessey Company to work on various applications of pattern recognition. He then joined BT to research OCR systems and led a team developing systems for the machine translation of text and speech. During this period he was also task leader on an ESPRIT project which developed equipment for the design validation of integrated circuits. More recently he managed a team designing spoken dialogues for new telephone services, such as CallMinder and Chargecard. From 1991 he managed one of BT's strategic university research initiatives at Edinburgh University aimed at producing new strategies for the evaluation of user interfaces to new telephone services. He now leads a group concerned with the development of strategies for content management in new multimedia services. He is a Corporate Member of the IEE and BCS.